

clustered index

=====

when you create a clustered index,

the records will be physically ordered in the actual table

order_id

you can have only one clustered index in a table.

orders..

order_id

customer_id

```
select order_id, order_customer_id from orders
where order_id = 5;
```

```
select order_id, order_customer_id from orders
where order_customer_id = 5;
```

```
create clustered index IX_orders_customer_id on orders (order_customer_id)
```

```
execute sp_helpindex orders;
```

```
SELECT
    i.name AS IndexName,
    i.type_desc AS IndexType,
    c.name AS ColumnName
FROM
    sys.indexes i
    INNER JOIN sys.index_columns ic ON i.object_id = ic.object_id AND
    i.index_id = ic.index_id
    INNER JOIN sys.columns c ON ic.object_id = c.object_id AND
    ic.column_id = c.column_id
WHERE
    i.object_id = OBJECT_ID('orders');
```

in this case I dropped the primary key

but in this case If I would have tried to drop the index only...

```
CREATE TABLE orders(  
  order_id int NOT NULL,  
  order_date datetime NOT NULL,  
  order_customer_id int NOT NULL,  
  order_status varchar(45) NOT NULL,  
  PRIMARY KEY (order_id)  
);
```

```
insert into orders  
select * from orders_temp;
```

```
drop index PK__orders__4659622911949DAC on orders;
```

```
alter table orders  
drop constraint PK__orders__4659622911949DAC;
```

```
create index IX_orders_customer_id_order_id_comp  
on orders (order_customer_id);
```

```
drop index IX_orders_customer_id_order_id_comp on orders;
```

```
create clustered index IX_orders_customer_id_order_id_comp on orders (order_customer_id);
```

```
create clustered index IX_orders_customer_id_order_id_comp on orders  
(order_customer_id,order_id);
```

```
-- I am using the first column as the predicate (seek)  
select * from orders  
where order_customer_id = 5;
```

```
-- I am using the second column as the predicate (scan)  
select * from orders  
where order_id = 5;
```

```
-- even if the order is reversed in where clause it does not matter  
select * from orders  
where order_id = 36472 and order_customer_id = 5;
```